

YEN-HSIU, CHEN

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Vision-Language Model Inference | Edge-Cloud Perception Systems | Real-Time Perception-to-Control Integration

SUMMARY

Ph.D. candidate in Computer Science at Yokohama National University, advised by Prof. Shinichi Shirakawa. My research centers on split inference and model compression, including quantization, feature compression, and hardware-aware architecture design, to run large vision and vision-language models under real hardware and bandwidth constraints, currently extended to LLaVA-series VLMs. My latest work is a bandwidth-aware split-inference framework, the only method that remains feasible under severe bandwidth degradation where standard splitting strategies miss the latency deadline entirely, cutting transmission by up to 99.73% while preserving up to 99.92% of original accuracy. Deployed on NVIDIA Jetson and Qualcomm NPUs. Integrated real-time perception with robotic control systems at CoLab. Embodied agents face these same constraints, limited onboard compute, strict latency budgets, and real-time perception and action.

EDUCATION

Yokohama National University

Apr 2022 – Mar 2027 (Expected)

Kanagawa, Japan | M.S. / Ph.D. in Computer Science and Information Engineering

- Graduate School of Environment and Information Sciences (Shirakawa Lab)
- (Fully-funded) Japan-Taiwan Exchange Association Scholarship covering both M.S. and Ph.D. degrees

PUBLICATIONS

- C. Yen-Hsiu, et al., “Adaptive Visual Token Selection and Mixed-Precision Quantization for Edge-Cloud Split Inference of Vision-Language Models,” *Target: IEEE CCNC 2027*. (Submitting)
- C. Yen-Hsiu, et al., “Residual Vector Quantization-Based Dynamic Bottleneck for Latency-Accuracy Trade-off in Split Inference,” *IEEE Internet of Things Journal*. (Under Review)
- C. Yen-Hsiu, et al., “Enhancing Latency-Accuracy Tradeoff in Dynamic Split Inference via Vector Quantized Bottleneck,” *IEEE GlobeCom Workshops*, 2024, pp. 1–6.
- S. Shimizu, . . . , C. Yen-Hsiu, et al., “Neural Architecture Search for Improving Latency-Accuracy Trade-off in Split Computing,” *IEEE GlobeCom Workshops*, 2022, pp. 1864–1870.

RELEVANT EXPERIENCE

AI Research & Development Intern

Aug 2022 – Oct 2024

CoLab Inc., Kanagawa Tools: Python, PyTorch, OpenCV, ROS, Linux, Docker

- **Real-Time Perception-to-Control Integration:** Engineered an inference-to-actuation pipeline on industrial cameras and embedded PCs, integrating deep learning perception with ROS and industrial PLC protocols to drive low-latency, safety-critical robotic control in a simulated factory environment.
- **End-to-End ML Pipeline & Internal Tooling:** Independently architected a full pipeline covering data collection, curation, model training, evaluation, optimization, and internal GUI tooling to let the team iterate rapidly, resulting in deployment into factory production.
- **Validation & Performance:** Achieved **97.5% detection accuracy** in a simulated factory environment and validated stability under continuous operational testing.

Model Acceleration & Algorithm Intern

Nov 2024 – Feb 2025

OPPO Inc., Kanagawa Tools: Python, PyTorch, Qualcomm SNPE/QNN SDK

- **NPU Quantization & Deployment Pipeline:** Built an end-to-end INT8/FP16 quantization pipeline for camera/vision models on Snapdragon NPU, including sensitivity analysis, conversion, and on-device validation for real-time edge deployment.
- **Research Evaluation:** Surveyed vision and quantization research to assess model size and accuracy trade-offs for mobile deployment feasibility, with results informing technical decisions.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, PyTorch, CUDA, C++
- **Languages:** Mandarin Chinese (Native), English (Proficient), Japanese (Proficient)